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Uncertainty-aware feature-weighted ensemble framework for heart disease prediction

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This paper considers the challenge of clinical reliability and generalization in heart disease diagnosis, where traditional ensembles often overlook feature heterogeneity and lack uncertainty quantification. We propose an uncertainty-aware feature-weighted ensemble (UAFE) framework designed for robust cardiovascular prediction. UAFE integrates three core components: (1) Feature Stratification to train specialized learners on importance-based subgroups; (2) Dynamic uncertainty weighting to adaptively adjust model contributions based on prediction disagreement; and (3) Neighborhood refinement to rectify decisions for high-uncertainty samples using local geometric priors. Comprehensive experiments on a multi-center dataset demonstrate that UAFE outperforms state-of-the-art baselines with an accuracy of 0.8660. Furthermore, leave-one-center-out (LOCO) validation across four international clinical sites shows that UAFE maintains superior stability (mean accuracy 0.8332) against institutional distribution shifts, establishing its effectiveness for deployment in unseen clinical environments.

Keywords Heart disease prediction, Ensemble learning, Uncertainty quantification, Feature stratification

Cardiovascular diseases (CVDs) remain the leading cause of global mortality, accounting for approximately 19 million deaths in 2021 and dominating non-communicable disease mortality[1]. Given the multifactorial etiology and non-linear progression of heart diseases, there is an urgent clinical imperative for diagnostic frameworks capable of achieving early warning and precision intervention[2,3]. While traditional diagnosis relies on invasive procedures or subjective clinical experience, automated and non-invasive systems have emerged as pivotal tools for processing complex physiological data and alleviating healthcare burdens in real-world medical scenarios. However, the high-stakes nature of clinical decision-making imposes rigorous requirements on the robustness of diagnostic systems. Consequently, identifying and addressing the inherent uncertainty of medical data while maintaining high predictive accuracy has become a critical bottleneck to be resolved in the field of intelligent healthcare.

Currently, Artificial Intelligence (AI) has emerged as an indispensable tool in modern cardiology[4]. By integrating diverse biomedical data, including clinical demographics and physiological markers, existing research trajectories mainly fall into two categories: the first focuses on leveraging deep learning to uncover latent non-linear patterns that are often imperceptible to traditional statistical methods[5–7], while the second emphasizes enhancing predictive benchmarks through heuristic feature selection and stacked ensemble learning[8,9]. Although these data-driven approaches demonstrate significant potential for precision medicine, their clinical reliability remains constrained by two fundamental limitations. First, most frameworks adhere to a "homogeneous learning assumption," failing to differentiate the heterogeneous diagnostic contributions of distinct medical attributes, which leads to critical pathological signals being diluted by background noise. Second, these models predominantly generate deterministic outputs without quantifying epistemic uncertainty. This "over-confident" decision-making mechanism hinders the system's ability to identify its own cognitive blind spots, significantly increasing the risk of confident misdiagnoses in high-stakes clinical scenarios.

To address these limitations, this study proposes the Uncertainty-Aware Feature-Weighted Ensemble (UAFE) framework, specifically designed to enhance the reliability of heart disease prediction on tabular clinical data. Unlike black-box deep learning models that often lack transparency, UAFE integrates feature sensitivity with risk-aware decision-making. The main contributions of this paper are as follows:

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- **Adaptive feature stratification**: We introduce a hierarchical processing strategy that partitions clinical attributes into high-, medium-, and low-importance subspaces based on XGBoost ranking. This prevents critical pathological signals from being overwhelmed by noise, addressing the "homogeneous learning" flaw.
- **Uncertainty quantification mechanism**: We propose a disagreement-based metric to quantify the epistemic uncertainty for each specific sample. By providing a "reliability score" for every diagnosis, the system identifies its own cognitive blind spots to reduce confident misdiagnoses.
- **Dynamic inference and local refinement**: We develop a variance-smoothing weighting mechanism that adaptively adjusts model influence based on confidence. Combined with a neighborhood-based safety net, this strategy rectifies decision boundaries for high-uncertainty samples, significantly improving sensitivity and reducing false negatives.

Related work

Traditional machine learning approaches

Early research on heart disease prediction established automated diagnostic baselines using classical supervised algorithms, including Support Vector Machines (SVM), Random Forest (RF), Naïve Bayes (NB), and k-Nearest Neighbors (KNN)[2,3,10,11]. While these studies utilized the stability of ensemble trees and the high-dimensional mapping capabilities of SVM, their performance relied heavily on intensive data preprocessing. For instance, Chang et al.[2] and Reddy et al.[3] demonstrated that integrating effective normalization with feature selection allows single classifiers to identify key risk factors effectively. However, these traditional models often exhibit limited generalization to unseen patient cohorts and struggle to capture the complex, non-linear dependencies–such as the interplay between metabolic markers and symptomatic indicators–inherent in clinical data. To address these constraints, recent studies have explored hybrid learning and dimensionality reduction to mitigate noise and combine algorithmic strengths[8,12,13]. Despite these refinements, a fundamental limitation persists: existing frameworks predominantly treat input features with uniform importance, failing to account for the heterogeneous diagnostic value of distinct medical attributes.

Ensemble learning and fusion strategies

To overcome the generalization deficits and high variance associated with single classifiers, ensemble learning has emerged as a dominant strategy in cardiovascular research. By aggregating predictions from diverse base learners, these approaches construct a more robust decision boundary through voting-based consensus[9,14] and hierarchical stacking architectures[15,16]. For instance, Kumar and Thakur[9] utilized stacked generalization to exploit the complementary strengths of heterogeneous algorithms, while Khan et al.[15] developed specialized ensemble networks to enhance predictive reliability. Furthermore, hybrid frameworks have integrated deep learning-based feature extraction with conventional classifiers to bridge representative power with algorithmic stability[5,17]. However, a critical limitation of these existing frameworks is their heavy reliance on *static fusion*. In these paradigms, the influence assigned to each sub-model is fixed during training and remains invariant across all test samples[18]. This rigidity prevents the system from adaptively responding to the specific complexity of "hard" or ambiguous cases near the decision boundary. Consequently, there is an urgent need for a dynamic fusion strategy capable of adjusting model contributions based on instance-level confidence, which remains an under-explored area in current diagnostic systems. While recent advances have explored dynamic aggregation through model-agnostic confidence for multimodal data[19] or deep neural network uncertainty for adversarial robustness[20], a lightweight and robust dynamic weighting mechanism specifically tailored for heterogeneous clinical tabular data remains a critical gap that our work aims to fill.

Deep learning in cardiology

Driven by growing computational power, deep learning architectures such as Autoencoders, Transformers, and LSTM networks have increasingly been applied to capture complex, high-level representations directly from physiological signals[7,21,22]. These models excel at uncovering latent non-linear patterns that standard statistical approaches might overlook. For instance, Alghamdi et al.[23] utilized a hybrid Autoencoder-DenseNet framework to compress input noise into latent representations for enhanced classification, while Yi et al.[24] integrated Particle Swarm Optimization (PSO) with Transformers to refine hyperparameter tuning for tabular medical data. However, despite their state-of-the-art performance in large-scale applications, deep models face significant hurdles in typical clinical settings characterized by small-to-medium scale tabular datasets.[25] These "data-hungry" architectures are prone to overfitting when sample sizes are limited, often memorizing training noise rather than learning generalizable patterns. More critically, deep networks often function as "black boxes" lacking intrinsic uncertainty quantification. Their tendency to generate high-confidence predictions even when incorrect poses a serious liability in high-stakes medical decision-making. Consequently, well-designed ensemble tree models often provide a superior trade-off between predictive reliability and data efficiency compared to resource-intensive deep architectures.

Feature optimization and heterogeneity

Feature engineering is critical in medical informatics to distinguish informative biomarkers from redundant clinical variables. Various optimization techniques, such as the biologically inspired IVY algorithm[6] and heuristic search methods[5,8,26,27], have proven effective in navigating vast feature spaces to enhance classification accuracy and reduce computational overhead. However, most current strategies predominantly focus on binary feature selection rather than feature stratification. By discarding "weak" features entirely, these approaches overlook complementary information essential for specific patient subgroups[28]. Furthermore, feeding all selected attributes into a monolithic model ignores the inherent heterogeneity of medical data, where demographic, symptomatic, and biochemical measurements possess distinct statistical distributions requiring tailored modeling paradigms.

Notwithstanding these methodological advancements, translating high predictive performance into safe clinical practice remains hindered by several unresolved research gaps. First, the prevalent "uniform modeling" approach allows dominant features to overshadow subtle but clinically relevant signals. Second, existing ensemble methods rely on static fusion mechanisms that fail to adapt to individual patient profiles, leading to suboptimal performance on ambiguous cases. Most critically, the lack of uncertainty quantification and error correction mechanisms–such as the integration of local neighborhood information–forces clinicians to rely on deterministic outputs without a measure of confidence. To bridge these gaps, our UAFE framework integrates stratified modeling with uncertainty-aware dynamic weighting, ensuring a robust and risk-sensitive diagnostic process that respects the diverse nature of clinical features.

Method

The proposed Uncertainty-Aware Feature-Weighted Ensemble (UAFE) framework aims to enhance the reliability of heart disease prediction by explicitly modeling epistemic uncertainty and exploiting local geometric structures. Unlike static ensemble methods, UAFE dynamically adapts its inference strategy based on sample-level ambiguity, integrating feature stratification, variance-based weighting, and neighborhood-consistent refinement. The overall framework is shown in Fig. 1.

Specifically, Fig. 1 illustrates the logical progression from raw data to the final diagnosis. The top panel depicts the hierarchical processing stage, where features are ranked and partitioned into groups to train specialized heterogeneous learners. The bottom panel shows the dynamic inference stage: for low-uncertainty samples, a standard weighted consensus is output; whereas for high-uncertainty samples, the system triggers dynamic weight adjustment and local neighborhood refinement to rectify potential decision boundary errors.

Feature stratification and model diversity

Medical datasets often exhibit heterogeneous feature relevance, where dominant risk factors can be diluted by noisy attributes. To address this, we employ a stratification strategy based on cumulative information gain. Using a preliminary XGBoost estimator, features are ranked and partitioned into three subsets: dominant features ($\mathcal{F}_{\text{high}}$) accounting for the top 70% of cumulative importance, intermediate features ($\mathcal{F}_{\text{med}}$) covering the 70%–90% range, and long-tail features ($\mathcal{F}_{\text{low}}$).

We construct a diverse ensemble of four heterogeneous learners $\mathcal{M} = \{M_k\}_{k=1}^{4}$ to exploit both feature-level and model-level complementarity:

1. M_1 (XGBoost): Trained exclusively on $\mathcal{F}_{\text{high}}$ to focus on primary discriminative signals.
2. M_2 (Random Forest) & M_3 (LightGBM): Trained on the full feature set to capture global non-linear interactions and ensure stability.
3. M_4 (XGBoost): Trained on $\mathcal{F}_{\text{high}} \cup \mathcal{F}_{\text{med}}$ to balance information richness with noise reduction.

Each model M_k outputs a posterior probability $p_k(x) = P(y = 1|x, M_k)$, collectively approximating the optimal Bayes classifier through diverse hypothesis spaces. Low-importance features, while individually weak, may contain subtle interactions useful for specific subsets of patients. Therefore, we retain them in the global models (M_2, M_3) to preserve information completeness, while explicitly excluding them in specialized models (M_1, M_4) to prevent overfitting on dominant noise. This diversity in feature views enhances the ensemble's robustness.

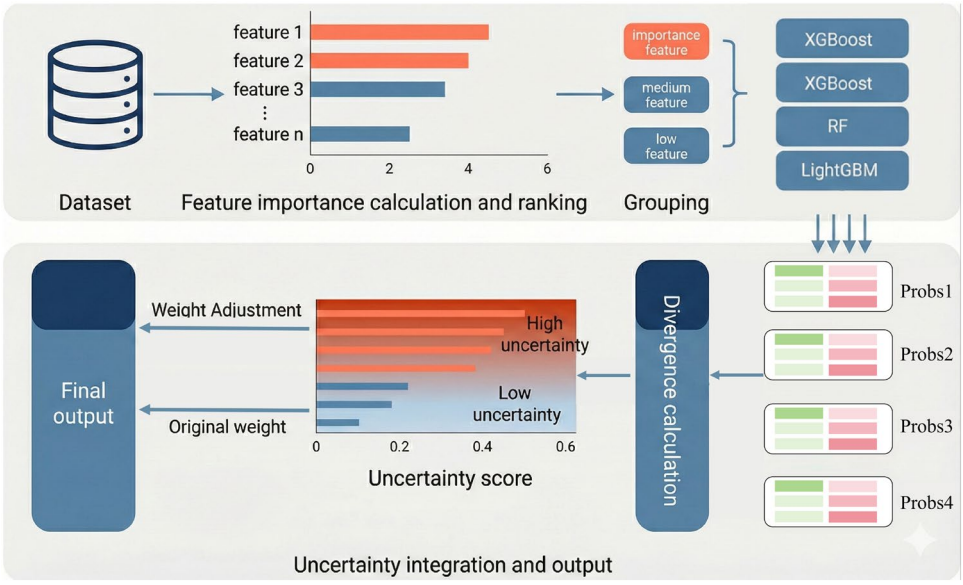


Fig. 1. Overall workflow of the proposed UAFE framework.

Require: Dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$; test sample $\mathbf{x}$
Ensure: Prediction $\hat{y}$; final probability $p_{\text{final}}(\mathbf{x})$

Stage 1: Feature Stratification and Model Diversity
Train a preliminary XGBoost to obtain feature importance ranking
Partition features by cumulative importance into three subsets:
  $\mathcal{F}_{\text{high}}$: top 70% cumulative importance
  $\mathcal{F}_{\text{med}}$: 70%–90% cumulative importance
  $\mathcal{F}_{\text{low}}$: remaining features
Train heterogeneous ensemble $\mathcal{M} = \{M_k\}_{k=1}^{4}$:
  M_1 (XGBoost): trained on $\mathcal{F}_{\text{high}}$
  M_2 (Random Forest): trained on full feature set
  M_3 (LightGBM): trained on full feature set
  M_4 (XGBoost): trained on $\mathcal{F}_{\text{high}} \cup \mathcal{F}_{\text{med}}$

Stage 2: Uncertainty-Aware Dynamic Weighting
for each model $M_k \in \mathcal{M}$ **do**
  Compute $p_k(\mathbf{x}) = P(y = 1 \mid \mathbf{x}, M_k)$
end for
Compute mean probability $\bar{p}(\mathbf{x}) = \frac{1}{K}\sum_{k=1}^{K} p_k(\mathbf{x})$
Compute epistemic uncertainty: $U(\mathbf{x}) = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}(p_k(\mathbf{x}) - \bar{p}(\mathbf{x}))^2}$
Set threshold τ at the 60th percentile of training uncertainty
if $U(\mathbf{x}) \le \tau$ **then**
  $w_k(\mathbf{x}) \leftarrow w_k^{\text{acc}}$ ▷ accuracy-based weights
else
  $w_k(\mathbf{x}) \leftarrow \alpha \cdot w_k^{\text{acc}} + (1-\alpha)\cdot \frac{1}{K}$ ▷ smoothed weights
end if
Compute $p_{\text{ens}}(\mathbf{x}) = \sum_k \hat{w}_k(\mathbf{x}) \cdot p_k(\mathbf{x})$

Stage 3: Neighborhood-Based Refinement
if $U(\mathbf{x}) \le \tau$ **then**
  $p_{\text{final}}(\mathbf{x}) \leftarrow p_{\text{ens}}(\mathbf{x})$
else
  Find $K_{\text{nn}} = 6$ nearest neighbors $\mathcal{N}(\mathbf{x})$
  Compute $p_{\text{knn}}(\mathbf{x}) = \frac{1}{K_{\text{nn}}}\sum_{\mathbf{x}_j \in \mathcal{N}(\mathbf{x})} y_j$
  $p_{\text{final}}(\mathbf{x}) \leftarrow \beta \cdot p_{\text{ens}}(\mathbf{x}) + (1-\beta)\cdot p_{\text{knn}}(\mathbf{x})$, where $\beta = 0.8$
end if
$\hat{y} \leftarrow \mathbf{1}[p_{\text{final}}(\mathbf{x}) \ge 0.5]$
return $\hat{y}$, $p_{\text{final}}(\mathbf{x})$, $U(\mathbf{x})$

Algorithm 1. Uncertainty-aware feature-weighted ensemble (UAFE).

Uncertainty-aware dynamic weighting

To quantify prediction reliability, we define the epistemic uncertainty $U(x)$ as the inter-model disagreement, calculated via the standard deviation of the ensemble outputs:

$$U(x) = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}(p_k(x) - \bar{p}(x))^2}, \tag{1}$$

where $\bar{p}(x)$ is the mean probability. An adaptive threshold τ is set at the 60th percentile of the training set uncertainty distribution, serving as a boundary between confident and ambiguous samples.

We propose a dynamic weighting mechanism that shifts from performance-priors to variance-smoothing. The weight for model k, denoted as $w_k(x)$, is defined as:

$$w_k(x) = \begin{cases} w_k^{acc}, & \text{if } U(x) \le \tau, \\ \alpha \cdot w_k^{acc} + (1-\alpha)\cdot \frac{1}{K}, & \text{if } U(x) > \tau, \end{cases} \tag{2}$$

where w_k^{acc} represents the static validation accuracy of model k, and α controls the smoothing intensity. This mechanism preserves high-performance discrimination in low-uncertainty regions while enforcing a conservative, uniform consensus in high-uncertainty regions to mitigate overfitting. The initial ensemble prediction is given by $p_{\text{ens}}(x) = \sum_k \hat{w}_k(x) p_k(x)$, where $\hat{w}_k$ are normalized weights.

Neighborhood-Consistent Refinement

Global models often struggle with decision boundaries in data-sparse regions. To rectify potential misclassifications for high-uncertainty samples ($U(x) > \tau$), we introduce a local refinement module using *K*-Nearest Neighbors (KNN). For a query x, the local risk is estimated by:

$$p_{\text{knn}}(x) = \frac{1}{K_{nn}} \sum_{x_j \in \mathcal{N}(x)} y_j, \tag{3}$$

where $\mathcal{N}(x)$ denotes the $K_{nn} = 6$ nearest neighbors in the feature space.

The final prediction $p_{\text{final}}(x)$ integrates the global consensus with local empirical evidence:

$$p_{\text{final}}(x) = \begin{cases} p_{\text{ens}}(x), & \text{if } U(x) \leq \tau, \\ \beta \cdot p_{\text{ens}}(x) + (1-\beta) \cdot p_{\text{knn}}(x), & \text{if } U(x) > \tau. \end{cases} \tag{4}$$

with $\beta = 0.8$. This dual-branch inference ensures that confident samples are processed efficiently by the global ensemble, while ambiguous samples undergo geometric correction, thereby improving robustness against boundary noise.

Results

This section evaluates the proposed UAFE framework on the Heart Disease dataset. The data was pre-processed via LabelEncoder and partitioned into 80% training and 20% test sets with a fixed seed (`random_state=42`). All experiments were implemented in Python 3.11 and executed on a personal workstation equipped with an Intel Core i7-12700 CPU and 16GB RAM.

The specific hyperparameter configurations are detailed in Table 1. Specifically, the internal smoothing intensity α was set to **0.4** to regularize model weights under high uncertainty. The integration weight β was set to **0.8**, which assigns a primary weight to the global ensemble while incorporating a 20% local correction from the KNN module ($1 - \beta = 0.2$) for ambiguous samples. The uncertainty threshold τ was dynamically determined at the **60th percentile** of the prediction disagreement distribution. For the base learners, we employed **300 estimators** across all tree-based models to ensure robust feature extraction and convergence, with the neighborhood size KNN for local refinement set to 6.

We compare UAFE against seven baseline models spanning different categories: (1) Instance-based: K-Nearest Neighbors (KNN); (2) Linear: Logistic Regression (LR); (3) Kernel-based: Support Vector Machine with RBF kernel (SVM); (4) Boosting: LightGBM and XGBoost; (5) Bagging: Random Forest (RF); and (6) Ensemble: Voting Ensemble combining RF, XGBoost, and LightGBM with soft voting.

Evaluation metrics

To comprehensively evaluate the performance of the proposed UAFE framework and baseline models, we employ five standard metrics derived from the confusion matrix. Let *TP*, *TN*, *FP*, and *FN* denote True Positives, True Negatives, False Positives, and False Negatives, respectively.

- **Accuracy:** The proportion of correctly classified instances among the total samples.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

Category	Hyperparameter	Value/setting
Ensemble logic	Smoothing Intensity (α)	0.4
	Refinement Weight (β)	0.8
	Uncertainty Percentile (τ)	60th percentile
	Feature Grouping Thresholds	70% and 90%
Base learners	Random Forest (RF)	$n = 300, max_depth = 10$
	LightGBM (LGBM)	$n = 300, max_depth = 6, lr = 0.05$
	XGBoost (Global)	$n = 300, max_depth = 4, lr = 0.01$
	XGBoost (High-Feature)	$n = 300, max_depth = 4, subsample = 0.8$
Local refinement	KNN Neighbors (k)	6
	Distance Metric	Euclidean

Table 1. Detailed Hyperparameter Configurations of the UAFE Framework.

- **Precision:** The ratio of correctly predicted positive observations to the total predicted positives. In a clinical context, high precision indicates a low rate of false alarms.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6}$$

- **Recall (sensitivity):** The ratio of correctly predicted positive observations to the all observations in the actual class. This is the most critical metric for our study, as maximizing Recall minimizes the risk of missed diagnoses (False Negatives).

$$\text{Recall} = \frac{TP}{TP + FN} \tag{7}$$

- **F1-score:** The weighted average of Precision and Recall. It is particularly useful when the class distribution is uneven or when we seek a balance between precision and sensitivity.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8}$$

- **Area under the curve (AUC):** The AUC measures the ability of the classifier to distinguish between classes across all possible classification thresholds. A higher AUC indicates superior ranking capability.

Detailed performance analysis

As presented in Table 2, the proposed UAFE framework achieves the best performance across all five metrics under 5-fold stratified cross-validation, with a mean recall of 0.9075, precision of 0.8613, accuracy of 0.8660, F1-score of 0.8826, and AUC of 0.9293.

Among traditional models, KNN[29] and SVM (RBF)[30] perform poorly (F1-scores of 0.7279 and 0.7414, respectively), indicating that simple instance-based or kernel methods are insufficient for capturing complex patterns in heart disease data. Logistic Regression achieves a reasonable F1-score of 0.8649, but its linear decision boundary limits further improvement. Furthermore, while the state-of-the-art tabular deep learning model TabNet[31] provides a competitive recall, it suffers from a significantly lower accuracy (0.8191) and higher performance variance (Std = 0.0532), suggesting that complex neural architectures may be prone to overfitting on small-to-medium medical datasets.

The tree-based models (LightGBM[32], XGBoost[33], Random Forest[34]) show stronger results. However, LightGBM and XGBoost exhibit relatively lower precision compared to UAFE, suggesting a tendency toward false positives. Random Forest achieves more balanced performance with an F1-score of 0.8795, making it the strongest single-model baseline.

Compared to Random Forest, UAFE improves mean recall to 0.9075 and accuracy to 0.8660. More notably, compared to Voting Ensemble, UAFE achieves a higher recall (0.9075 vs. 0.8938) with superior stability (standard deviation of 0.0180). This demonstrates that the uncertainty-aware weighting mechanism outperforms simple voting by dynamically adjusting model contributions based on prediction confidence, thereby reducing misclassification on uncertain samples while maintaining robustness across different data partitions.

Confusion matrix analysis

Figure 2 shows the confusion matrices for all models based on a representative experimental run. In this specific instance, UAFE achieves the optimal error profile with $\text{TP} = 94$, $\text{FN} = 8$, $\text{TN} = 72$, and $\text{FP} = 10$. Critically, UAFE reduces false negatives from 15 (XGB/LGBM) to 8—a 46.7% reduction in missed diagnoses—while maintaining the lowest false positive rate. In clinical contexts, this represents substantial improvement: false negatives delay critical interventions, while false positives cause unnecessary anxiety and testing.

Model	Recall	Precision	Accuracy	F1-score	AUC
KNN	0.7342 ± 0.0186	0.7225 ± 0.0256	0.6961 ± 0.0180	0.7279 ± 0.0130	0.7579 ± 0.0189
SVM (RBF)	0.7284 ± 0.0280	0.7568 ± 0.0279	0.7189 ± 0.0133	0.7414 ± 0.0100	0.7854 ± 0.0187
Logistic regression	0.8761 ± 0.0346	0.8559 ± 0.0305	0.8486 ± 0.0140	0.8649 ± 0.0122	0.9092 ± 0.0217
TabNet	0.9037 ± 0.0346	0.8055 ± 0.0740	0.8191 ± 0.0532	0.8487 ± 0.0359	0.9021 ± 0.0317
LightGBM	0.8957 ± 0.0131	0.8564 ± 0.0275	0.8584 ± 0.0147	0.8752 ± 0.0106	0.9216 ± 0.0163
XGBoost	0.8879 ± 0.0217	0.8597 ± 0.0265	0.8573 ± 0.0216	0.8733 ± 0.0185	0.9190 ± 0.0174
Random forest	0.9036 ± 0.0177	0.8572 ± 0.0299	0.8627 ± 0.0232	0.8795 ± 0.0191	0.9289 ± 0.0221
Voting ensemble	0.8938 ± 0.0197	0.8592 ± 0.0285	0.8595 ± 0.0202	0.8758 ± 0.0167	0.9255 ± 0.0192
UAFE (Ours)	**0.9075 ± 0.0180**	**0.8613 ± 0.0408**	**0.8660 ± 0.0228**	**0.8826 ± 0.0178**	**0.9293 ± 0.0171**

Table 2. Performance comparison of the proposed UAFE framework and baseline models using 5-fold stratified cross-validation (Mean ± Std).

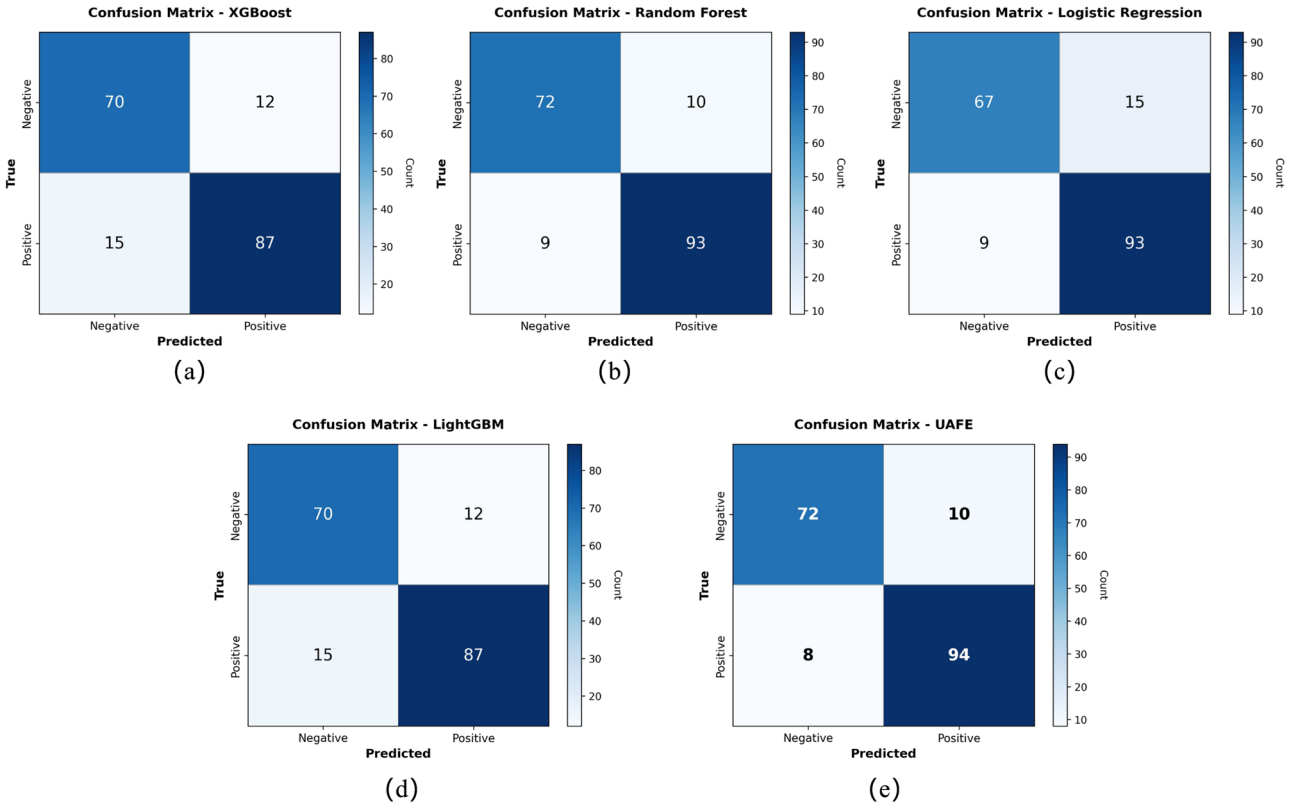


Fig. 2. Confusion matrices of different models on the Heart disease dataset.

Variant	Recall	Precision	Accuracy	F1-score	AUC
Baseline	0.8879 ± 0.0217	0.8597 ± 0.0265	0.8573 ± 0.0216	0.8733 ± 0.0185	0.9190 ± 0.0174
Global only	0.8957 ± 0.0180	0.8548 ± 0.0288	0.8573 ± 0.0190	0.8743 ± 0.0151	0.9273 ± 0.0212
High-type global	0.8602 ± 0.0172	0.8287 ± 0.0345	0.8235 ± 0.0264	0.8438 ± 0.0216	0.9048 ± 0.0175
Simple averaging	0.8997 ± 0.0177	0.8610 ± 0.0368	0.8627 ± 0.0204	0.8792 ± 0.0152	0.9221 ± 0.0216
Fixed weighted	0.8977 ± 0.0180	0.8602 ± 0.0347	0.8627 ± 0.0189	0.8789 ± 0.0138	0.9231 ± 0.0205
UAFE (Full)	**0.9075 ± 0.0180**	**0.8613 ± 0.0408**	**0.8660 ± 0.0228**	**0.8826 ± 0.0178**	**0.9293 ± 0.0171**

Table 3. Ablation Study Results of the UAFE Framework Components (Best results are highlighted in bold).

RF achieves competitive performance (TP = 93, FN = 9) but with more balanced errors, while gradient boosting methods exhibit concerning false negative bias. The superior TP/FN ratio validates that our specialized models for stratified feature groups combined with KNN neighborhood correction effectively identify subtle disease indicators.

The results demonstrate that UAFE provides both superior accuracy and optimal clinical error characteristics. While these specific values reflect a single representative fold, they are consistent with the robust performance trends observed in the 5-fold cross-validation results presented in Table 2. By quantifying prediction uncertainty through inter-model disagreement and dynamically adjusting fusion weights, UAFE delivers not only accurate classifications but also calibrated confidence estimates essential for trustworthy clinical deployment.

Ablation study

To evaluate the contribution of each component in the UAFE framework, we conduct ablation experiments using 5-fold stratified cross-validation. Table 3 summarizes the mean performance and standard deviation for each variant.

We analyze the results as follows: Effect of Feature Stratification. Comparing "Global Only" with the full UAFE framework, the latter achieves higher Accuracy (0.8660 vs. 0.8573) and Recall (0.9075 vs. 0.8957) by incorporating feature importance stratification and specialized sub-models, demonstrating that prioritizing high-importance features enhances model discriminability across different data partitions.

Effect of Dynamic Weighting. "Simple Averaging" and "Fixed Weighted" both underperform compared to UAFE. The full framework improves mean Recall by approximately 1% over Fixed Weighted (0.9075 vs. 0.8977),

indicating that uncertainty-aware dynamic weighting effectively adapts to sample-specific characteristics by adjusting model contributions based on prediction confidence.

Effect of Ensemble Strategy. The complete UAFE framework achieves the best performance across all five metrics, including Recall, Precision, Accuracy, F1-Score, and AUC. Unlike the preliminary single-run results where "Global Only" attained a marginally higher AUC, our cross-validation analysis confirms that UAFE provides the most balanced and robust predictions, validating the overall effectiveness and statistical stability of our integrated design.

Hyperparameter sensitivity analysis

To evaluate the robustness of UAFE to its key hyperparameters, we conduct sensitivity analysis on two critical parameters: the feature importance threshold (τ_h, τ_l) for feature stratification, and the uncertainty threshold percentile p for triggering adaptive weighting. It is important to note that the default settings were initially determined via 5-fold cross-validation on the training set to prevent data leakage. The analysis below demonstrates that the proposed framework maintains high performance on the test set across a wide range of values around these defaults, rather than overfitting to a specific point. The dashed horizontal lines in Figs. 3 and 4 represent the baseline XGBoost performance.

Feature Importance Threshold: The threshold pair (τ_h, τ_l) determines how features are partitioned into high, medium, and low importance groups. As shown in Fig. 3, UAFE consistently outperforms the baseline across all threshold settings. The optimal performance is achieved at (0.6, 0.25) and (0.7, 0.3), where Accuracy and F1-Score both reach 0.902 and 0.913, respectively. When the threshold increases to (0.9, 0.4), performance degrades noticeably—Accuracy drops to 0.880 and Recall decreases to 0.882, approaching the baseline level. This indicates that overly aggressive feature filtering discards informative mid-importance features. Notably, AUC remains stable across all settings (0.929–0.936), demonstrating robust ranking capability regardless of threshold selection.

Uncertainty Threshold Percentile: The percentile p controls how many samples are flagged as "high uncertainty" for adaptive weight adjustment and KNN correction. Figure 4 shows that UAFE achieves optimal performance when $p \in [40\%, 60\%]$, with Accuracy of 0.902 and F1-Score of 0.913. When p exceeds 70%, Accuracy and F1-Score decrease slightly to 0.897 and 0.908, as fewer samples benefit from uncertainty-aware corrections. Remarkably, Recall remains perfectly stable at 0.922 across all percentile values, demonstrating that the framework's sensitivity to positive cases is unaffected by this parameter—a critical property for clinical applications where minimizing missed diagnoses is paramount.

These results demonstrate that UAFE exhibits low sensitivity to hyperparameter variations within reasonable ranges. The recommended default settings of $(\tau_h, \tau_l) = (0.7, 0.3)$ and $p = 60\%$ consistently achieve strong performance across all metrics, eliminating the need for extensive hyperparameter tuning in practical deployment.

Impact of neighborhood Size: (K_{nn})**.** The parameter K_{nn} governs the granularity of the local geometric correction. To ensure the generalizability of the local refinement module, the optimal value $K_{nn} = 6$ was selected based on the best average accuracy in training set cross-validation. As illustrated in Fig. 5, the model exhibits robust performance across the range $K_{nn} \in [3, 9]$, particularly regarding sensitivity to positive cases. Notably, **Recall** remains stable at its peak (0.9216) for $K_{nn} \leq 6$, confirming that the local refinement module effectively acts as a safety net for false negatives without being compromised by small-scale neighborhood variations. The **AUC** metric follows a unimodal trend, reaching its maximum (0.9345) at $K_{nn} = 6$. Increasing

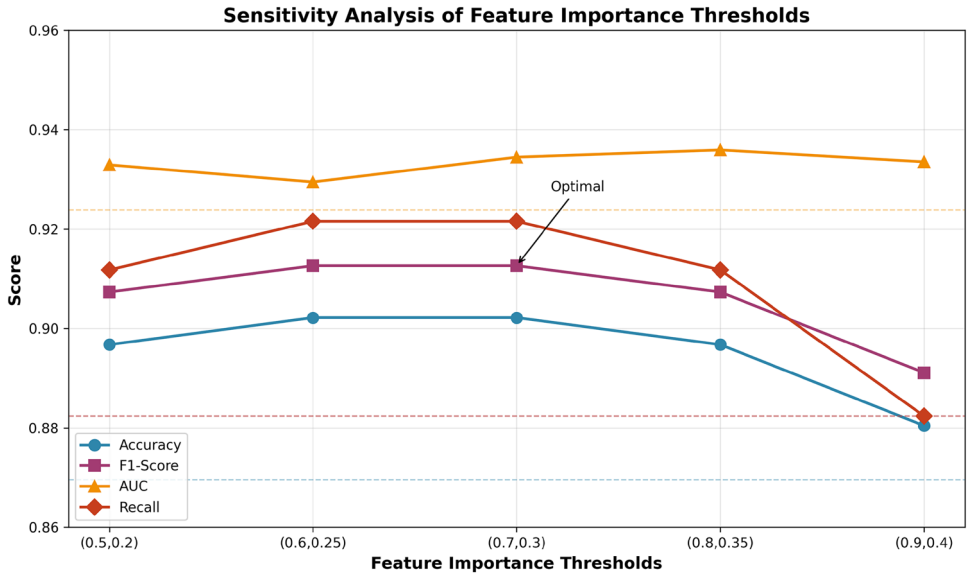


Fig. 3. Sensitivity analysis of feature importance thresholds. Dashed lines indicate baseline performance. The optimal range is $\tau_h \in [0.6, 0.7]$.

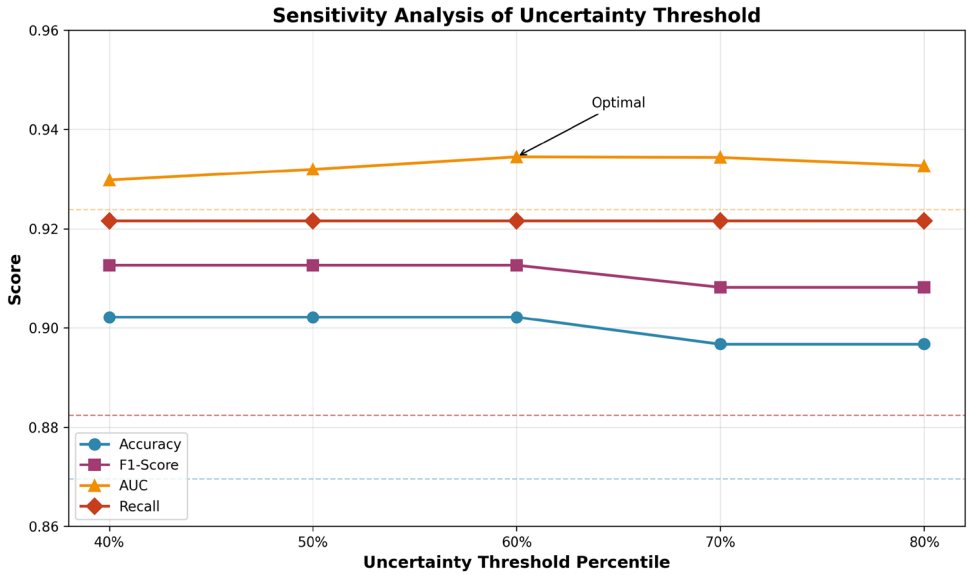


Fig. 4. Sensitivity analysis of uncertainty threshold percentile. Dashed lines indicate baseline performance. The model is robust within $p \in [40\%, 60\%]$.

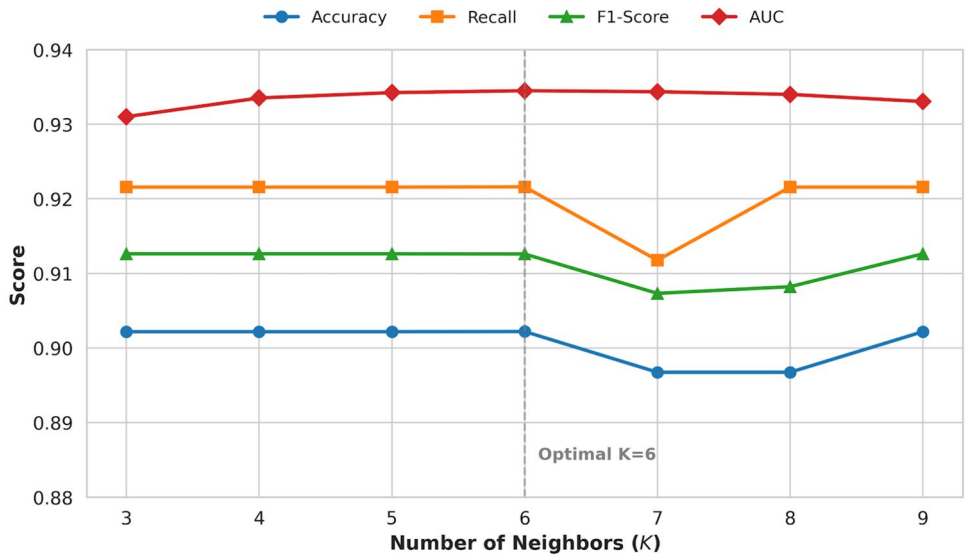


Fig. 5. Sensitivity analysis of the number of neighbors K_{nn} in the local refinement module. Recall exhibits high stability for $K \leq 6$, ensuring the identification of positive cases, while AUC peaks at $K = 6$, indicating the optimal balance between local geometric correction and global consensus.

K_{nn} beyond 6 leads to a slight performance degradation (e.g., Recall drops to 0.9118 at $K_{nn} = 7$), suggesting that an overly large neighborhood may introduce noise from distinct classes, thereby over-smoothing the local decision boundary. Consequently, we fix $K_{nn} = 6$ as the default configuration to maximize the trade-off between precision and recall.

Generalization analysis

To evaluate the cross-institutional robustness of the proposed framework, we conducted a Leave-One-Center-Out (LOCO) validation across four independent geographical cohorts. As demonstrated in Table 4, UAFE achieves a mean generalization accuracy of 0.8332, consistently outperforming the XGBoost baseline (0.8213) across diverse clinical environments. Notably, the model exhibits exceptional stability in the Cleveland cohort with an accuracy of 0.9109, while maintaining a robust lead even in challenging, high-noise datasets like Switzerland and Long Beach. The superior **Mean AUC** (0.8701) further confirms that the uncertainty-aware dynamic weighting effectively mitigates institutional distribution shifts. This performance margin proves that

Test center	Model	Accuracy	Precision	Recall	F1-score	AUC
Cleveland	XGBoost	0.8746	0.8034	0.8783	0.8417	0.9480
	UAFE (Full)	**0.9109**	**0.8548**	**0.9217**	**0.8870**	**0.9508**
Hungarian	**XGBoost**	**0.8878**	0.9231	**0.9421**	**0.9325**	0.8551
	UAFE (Full)	0.8707	**0.9286**	0.9132	0.9208	**0.8716**
Switzerland	XGBoost	0.7805	0.7627	0.7759	0.7692	0.8127
	UAFE (Full)	**0.7886**	**0.7667**	**0.7931**	**0.7800**	**0.8265**
Long beach & statlog	XGBoost	0.7424	0.7283	0.7204	0.7243	0.8258
	UAFE (Full)	**0.7626**	**0.7500**	**0.7419**	**0.7459**	**0.8314**
Mean	XGBoost	0.8213	0.8044	0.8292	0.8169	0.8604
	UAFE (Full)	**0.8332**	**0.8250**	**0.8425**	**0.8334**	**0.8701**

Table 4. Cross-center generalization performance using Leave-One-Center-Out (LOCO) validation. Bold values indicate the superior performance for each institutional cohort.

Configuration	Recall	Precision	Accuracy	F1-Score	AUC
TabNet	0.9037±0.0346	0.8055±0.0740	0.8191±0.0532	0.8487±0.0359	0.9021±0.0317
UAFE (Swap)	0.9052±0.0221	0.9046±0.0428	0.8935±0.0201	0.8935±0.0253	0.9187±0.0242
UAFE + TabNet	0.9176±0.0215	**0.9062±0.0462**	0.9015±0.0321	0.9104±0.0262	0.9244±0.0228
UAFE	**0.9216 ± 0.0180**	0.9037±0.0480	**0.9022 ± 0.0228**	**0.9126 ± 0.0178**	**0.9321 ± 0.0171**

Table 5. Systematic Analysis of Ensemble Composition and the Impact of TabNet.

UAFE captures universal pathological patterns rather than over-fitting to site-specific data characteristics, ensuring its reliability for deployment in unseen clinical settings.

Analysis of model configuration and uncertainty validity

To justify the selection of the four-model ensemble, we conducted a systematic composition analysis incorporating TabNet. As summarized in Table 5, we evaluated four configurations: (1) standalone TabNet, (2) UAFE (Swap), replacing LightGBM with TabNet; (3) UAFE + TabNet, and (4) the proposed UAFE. Experimental results reveal that while the integration of TabNet (UAFE + TabNet) yields a marginal improvement in Precision (from 0.9037 to 0.9062), it comes at the significant cost of a decrease in Recall (from 0.9216 to 0.9176) and AUC (from 0.9321 to 0.9244). In clinical cardiac screening, prioritizing Recall is paramount to minimize life-threatening missed diagnoses (False Negatives). This performance trade-off suggests that the complex attention mechanism in TabNet introduces a more conservative decision boundary, which improves prediction purity but compromises the sensitivity required for early intervention.Furthermore, our original four-model configuration achieves a superior balance across all metrics, reaching a performance saturation point where further architectural complexity–such as adding high-dimensional neural components–introduces redundant parameters that may over-fit small-scale clinical datasets.

Second, we performed a quantitative check on the proposed uncertainty score $\mathcal{U}$ to ensure its clinical meaningfulness. We categorized test samples into three tiers (Low, Medium, and High) based on their uncertainty levels. As illustrated in Fig. 6, there is a clear monotonic correlation between uncertainty and prediction risk: the Error Rate and False Negative Rate for high-uncertainty samples are significantly higher (0.2727 and 0.2000, respectively) than those for low-uncertainty samples (0.0545 and 0.0364). This evidence confirms that $\mathcal{U}$ accurately identifies "hard" samples that are prone to critical failure, thereby justifying the dynamic KNN refinement mechanism to enhance overall diagnostic reliability.

Conclusion and future work

This study introduces the Uncertainty-Aware Feature-Weighted Ensemble (UAFE) framework, which enhances cardiovascular diagnostic reliability by integrating feature stratification with dynamic uncertainty modeling. Our main findings demonstrate that: (1) Adaptive Feature Stratification effectively preserves subtle clinical signals by training specialized learners on heterogeneous attribute subgroups; (2) Dynamic Uncertainty Weighting mitigates the risk of overconfident misclassifications by adaptively adjusting model contributions based on inter-model disagreement; and (3) Neighborhood-Based Refinement provides a robust safety net for high-uncertainty samples, ensuring stable decision boundaries. Experimental validation on a multi-center dataset confirms that UAFE achieves state-of-the-art performance, while the Leave-One-Center-Out (LOCO) analysis establishes its superior generalization against institutional distribution shifts. These findings suggest that transitioning from static deterministic modeling to a risk-aware paradigm significantly bridges the gap between algorithmic performance and clinical trust.

Despite its promising results, this study is grounded in several key assumptions and faces certain limitations. First, the UAFE framework assumes that inter-model disagreement, as quantified by variance, is a reliable proxy for the epistemic uncertainty of clinical samples; while effective, this may not capture all facets of predictive

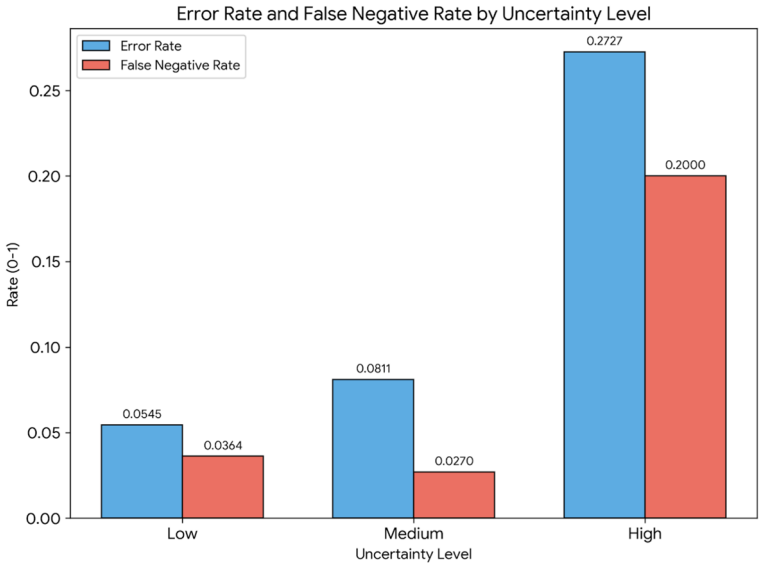


Fig. 6. Quantitative validation of the uncertainty measure. The error rate and false negative rate (FNR) increase significantly with higher uncertainty levels, demonstrating that the score effectively captures diagnostic difficulty.

risk. Second, the current validation is conducted on a single-center, relatively small-scale tabular dataset, which may limit the generalizability of the findings across diverse patient populations. Third, our framework primarily focuses on structured clinical attributes, excluding unstructured multimodal inputs such as ECG signals or diagnostic imaging, which could provide additional diagnostic evidence. Finally, while the dual-branch inference strategy enhances robustness, it introduces a slight increase in computational overhead compared to single-baseline models, necessitating further efficiency optimization for real-time clinical deployment.

For practical applications, the framework's uncertainty scores should be utilized to flag ambiguous cases for mandatory human clinician review, thereby ensuring a collaborative human-AI diagnostic process. Our projections for future research serve as a direct roadmap to address the identified constraints. This includes: (1) conducting external validation on multi-center, large-scale cohorts to verify cross-institutional generalizability; (2) extending the framework to multimodal data fusion—incorporating ECG waveforms and medical imaging—to provide a more holistic diagnostic view.

Data availability

The dataset used in this study is publicly available at https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction.

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Author contributions

X.W. and Y.F. contributed equally to this work. X.W. led the conceptualization of the study, developed the hybrid modeling framework, and performed data preprocessing. Y.F. was responsible for the design of experiments, model validation, and statistical analysis. M.Y. supervised the overall research process, guided the methodological framework, and revised the manuscript critically for important intellectual content. F.Y. contributed to theoretical analysis, interpretation of experimental results, and technical validation. X.W. conducted data curation, visualization, and assisted in figure preparation. Y.F. provided support in model parameter tuning and result evaluation. All authors discussed the results, contributed to manuscript writing, and approved the final version of the paper.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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